Light

Cyber Security — Prevention

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AQA GCSE Computer Science — **Cyber Security: Detection and Prevention (8525 Paper 2)**. This topic covers the definition and purpose of cyber security, and the methods used to detect and prevent cyber security threats including biometric measures, password systems, CAPTCHA, email confirmations, and automatic software updates. Exam questions typically ask you to **define** cyber security, **describe** prevention methods, **explain** how security measures protect against specific threats, and **discuss** the advantages and disadvantages of different approaches. Questions range from 1-mark definitions to 6-mark extended responses.

Why cyber security matters

Every day, individuals and organisations face threats from hackers, malware, and social engineering attacks. **Cyber security** is about protecting computer systems, networks, and data from these threats. Without proper security measures, personal information can be stolen, systems can be damaged, and businesses can lose money and reputation.

Understanding how to detect and prevent cyber attacks is essential in the modern digital world. The security measures covered in this topic are used everywhere —

from unlocking your phone with a fingerprint to proving you're human when signing up for a website.

What you must know

This checklist covers the key exam-visible concepts from the AQA specification. You should be able to explain each point clearly.

- **Define** the term cyber security.
- **Describe** the main purposes of cyber security.
- **Explain** how biometric measures can be used to protect systems, particularly for mobile devices.
- **Explain** how password systems protect against unauthorised access.
- **Explain** what CAPTCHA is and how it helps prevent automated attacks.
- **Explain** how email confirmations can be used to verify a user's identity.
- **Explain** why automatic software updates are important for security.
- **Understand** how different security measures can be combined to provide stronger protection.
- **Describe** how to protect against social engineering, malware, and other cyber threats.

> **Specification note.** AQA 8525 section 3.6.3 focuses on five specific prevention methods: biometric measures, password systems, CAPTCHA, email confirmations, and automatic software updates. You should understand what each method does and when it would be appropriate to use it. Biometric measures are particularly relevant for mobile devices.

Toolkit

The five prevention methods

AQA requires you to understand these five specific security measures:

Biometrics
Fingerprint, face, iris

Passwords
Secret credentials

CAPTCHA
Human verification

Email confirmations
Identity verification

Auto updates
Patch vulnerabilities

Prevention methods at a glance

Method	What it does	Protects against
Biometric measures	Uses unique physical characteristics to verify identity	Unauthorised access; password theft
Password systems	Requires knowledge of a secret to gain access	Unauthorised access
CAPTCHA	Tests that prove user is human, not a bot	Automated attacks; spam; brute force
Email confirmations	Verifies identity by sending a link/code to registered email	Account takeover; fake accounts
Automatic software updates	Patches security vulnerabilities without user action	Exploits of known vulnerabilities

Key definitions

Term	Definition (exam answer)
Cyber security	The processes, practices and technologies designed to protect networks, computers, programs and data from attack, damage or unauthorised access
Biometrics	Using unique physical characteristics (fingerprint, face, iris) to verify identity
CAPTCHA	A test designed to distinguish humans from automated bots
Patch	A software update that fixes security vulnerabilities or bugs

Threat-prevention matching

Threat	Prevention methods
Weak/default passwords	Strong password policies; biometrics
Phishing attacks	Email confirmations; user training
Automated attacks (bots)	CAPTCHA
Unpatched software	Automatic software updates

Threat	Prevention methods
Shoulder surfing	Biometrics (no visible password to observe)
Malware exploiting vulnerabilities	Automatic software updates; anti-malware

1. What is cyber security?

Cyber security is the protection of computer systems and networks from digital threats. It encompasses all the measures taken to keep data safe, prevent unauthorised access, and ensure systems continue to function correctly.

1.1 Definition of cyber security

Definition: Cyber security consists of the processes, practices and technologies designed to protect networks, computers, programs and data from attack, damage or unauthorised access.

This definition from the AQA specification has three key components:

- **Processes:** Procedures and policies that organisations follow (e.g., regular password changes, security training)
- **Practices:** Day-to-day behaviours that maintain security (e.g., not sharing passwords, checking email senders)
- **Technologies:** Tools and systems that provide protection (e.g., firewalls, encryption, biometrics)

1.2 Purposes of cyber security

Cyber security serves several important purposes:

- **Protecting data:** Keeping personal, financial, and confidential information safe from theft or exposure
- **Preventing unauthorised access:** Ensuring only authorised users can access systems and data
- **Maintaining system availability:** Keeping systems running and accessible when needed
- **Preventing damage:** Stopping malware and attacks from corrupting or destroying data
- **Protecting privacy:** Ensuring personal information is not exposed or misused
- **Maintaining trust:** Ensuring customers and users can trust organisations with their data

1.3 Why prevention matters

Prevention is better than cure in cyber security. Once data has been stolen or systems have been compromised, the damage is often already done. Prevention methods aim to:

- **Stop attacks before they succeed:** Block threats at the point of entry
- **Verify user identity:** Ensure the person accessing a system is who they claim to be
- **Close vulnerabilities:** Fix weaknesses before attackers can exploit them
- **Detect suspicious activity:** Identify potential threats early

Worked Example 1 — Defining cyber security

Question: Define the term cyber security. [2 marks]

Model answer:
Cyber security is the processes, practices and technologies **(1)** designed to protect networks, computers, programs and data from attack, damage or unauthorised access **(1)**.

Exam tip: Learn the full definition — both parts are needed for full marks. Remember the three Ps: Processes, Practices, and Protection technologies.

Worked Example 2 – Purposes of cyber security

Question: State two purposes of cyber security. [2 marks]

Model answer:
1. To protect data from theft or unauthorised access **(1)**
2. To prevent damage to computer systems from malware or attacks **(1)**

Exam tip: Make sure your two purposes are clearly different. Other valid answers include: maintaining system availability, protecting privacy, preventing financial loss.

2. Prevention methods

AQA requires you to understand five specific methods for detecting and preventing cyber security threats. Each method has different strengths and is appropriate for different situations.

2.1 Biometric measures

Biometric measures use unique physical characteristics to verify a person's identity. Unlike passwords, biometric data cannot be forgotten, guessed, or easily shared.

Definition: Biometric measures use unique physical characteristics of an individual to verify their identity.

Common types of biometric measures include:

- **Fingerprint recognition:** Scans the unique pattern of ridges on a finger
- **Facial recognition:** Analyses facial features such as the distance between eyes, nose shape, and jawline to identify a person
- **Iris/retina scanning:** Scans the unique patterns in the coloured part of the eye (iris) or blood vessels at the back of the eye (retina)
- **Voice recognition:** Identifies a person by analysing their voice patterns and speech characteristics

How biometric authentication works

Biometric systems work in two stages:

1. **Enrolment:** The user's biometric data (e.g., fingerprint) is captured and stored as a template. This happens once when setting up the device or account.
2. **Verification:** Each time the user wants to access the system, their biometric data is captured again and compared to the stored template. If it matches, access is granted.

Biometrics and mobile devices

Biometrics are **particularly useful for mobile devices** where typing long passwords is inconvenient. The AQA specification specifically highlights this. Modern smartphones commonly use:

- **Fingerprint sensors:** Built into the screen or a button — the user simply touches to unlock
- **Facial recognition:** The front camera scans the user's face — the user just looks at the phone to unlock

These methods are faster than typing a password and work well on small touchscreens. They also prevent "shoulder surfing" because there is no password for an observer to see.

Advantages of biometrics

- Cannot be forgotten (unlike passwords)
- Difficult to fake or steal
- Quick and convenient to use
- Cannot be shared with others
- No need to remember complex passwords

Disadvantages of biometrics

- Expensive hardware required
- Can fail (dirty fingers, poor lighting)
- If compromised, cannot be changed
- Privacy concerns about storing biometric data
- May not work for all users

2.2 Password systems

Password systems are the most common form of authentication. Users must enter a secret password that matches a stored value to gain access.

How password authentication works

When you create an account with a password:

1. **Password creation:** You choose a password, which is stored by the system (usually in encrypted form)
2. **Login attempt:** When you try to log in, you enter your password
3. **Comparison:** The system compares what you entered to the stored password
4. **Access decision:** If they match, you are granted access; if not, access is denied

Features of effective password systems

Good password systems don't just check if a password is correct — they include features to make passwords more secure:

- **Minimum length requirements:** Passwords must be at least 8-12 characters — longer passwords are harder to guess

- **Complexity requirements:** Must include uppercase, lowercase, numbers, and symbols — increases the number of possible combinations
- **Password history:** Prevents reusing recent passwords — stops users cycling through the same weak passwords
- **Account lockout:** Locks account after several failed attempts — prevents attackers trying thousands of guesses
- **Regular password changes:** Forces users to update passwords periodically — limits the time a stolen password can be used

What makes a strong password?

A strong password should be:

- **Long:** At least 8-12 characters
- **Complex:** Mix of uppercase, lowercase, numbers, and symbols
- **Unique:** Different for each account
- **Not personal:** Avoid names, birthdays, or common words
- **Not obvious:** Avoid "password", "123456", or keyboard patterns

Advantages of passwords

- No special hardware needed
- Can be changed if compromised
- Familiar to all users
- Works on any device
- Low cost to implement

Disadvantages of passwords

- Can be forgotten
- Can be guessed or cracked
- Users often choose weak passwords
- Can be observed (shoulder surfing)
- Can be shared with others

2.3 CAPTCHA

CAPTCHA stands for "Completely Automated Public Turing test to tell Computers and Humans Apart". It is a test designed to determine whether the user is a human or an automated bot.

Definition: CAPTCHA is a test designed to distinguish between human users and automated bots/programs.

How CAPTCHA works

CAPTCHA works by presenting a challenge that is easy for humans to solve but difficult for computer programs. The idea is based on the "Turing test" — if a computer cannot

tell whether the response came from a human or a machine, then CAPTCHA assumes it was a human.

Common types of CAPTCHA

- **Distorted text:** Users must type letters/numbers shown in a distorted, warped image. Computers struggle to read distorted text, but humans can usually work it out.
- **Image selection:** Users must select all images containing a specific object (e.g., "select all traffic lights" or "click on all the bicycles"). This requires visual understanding that computers find difficult.
- **Checkbox CAPTCHA:** Users click "I'm not a robot" — the system analyses mouse movement patterns, timing, and browsing behaviour to determine if the user is human.
- **Audio CAPTCHA:** Users listen to and type spoken letters/numbers. This provides accessibility for users with visual impairments.

What CAPTCHA protects against

CAPTCHA is designed to stop automated programs, not to verify identity. It protects against:

- **Brute force attacks:** Bots trying thousands of password combinations — CAPTCHA slows them down or blocks them completely
- **Spam:** Automated programs creating fake accounts or posting spam comments
- **Ticket scalping:** Bots buying all available tickets before humans can purchase them
- **Web scraping:** Automated programs harvesting data from websites
- **Fake reviews/votes:** Bots creating fake reviews or voting multiple times

Advantages of CAPTCHA

- Blocks automated attacks effectively
- No user account required
- Widely used and understood
- Prevents spam and fake accounts

Disadvantages of CAPTCHA

- Can be frustrating for users
- Accessibility issues for some users
- Advanced bots can sometimes solve them
- Takes extra time to complete

2.4 Email confirmations

Email confirmations verify a user's identity by sending a message to their registered email address. The user must click a link or enter a code to prove they have access to that email account.

Purpose: Email confirmations verify that the user has access to the email address they claim to own, confirming their identity.

How email confirmations work

The process typically works like this:

1. **Action triggered:** A user performs an action that requires verification (e.g., registering, resetting password, making a payment)
2. **Email sent:** The system sends an email to the registered address containing a unique link or code
3. **User responds:** The user clicks the link or enters the code on the website
4. **Verification complete:** The system confirms the user has access to that email, so they are likely genuine

Common uses of email confirmations

- **Account registration:** Confirming the email address is real and belongs to the user — prevents fake accounts
- **Password resets:** Verifying identity before allowing a password change — prevents account takeover
- **Suspicious activity alerts:** Asking the user to confirm they initiated an unusual action (e.g., login from new location)
- **Important changes:** Verifying changes to account settings, payment details, or personal information
- **Transaction confirmations:** Confirming large purchases or money transfers

Advantages of email confirmations

- Verifies the email address is valid
- No special hardware needed
- Provides an audit trail
- User already has access to email

Disadvantages of email confirmations

- Relies on email account being secure
- Emails can be delayed or go to spam
- If email is compromised, so is verification
- Requires internet access to check email

2.5 Automatic software updates

Automatic software updates ensure that security patches are installed without requiring user action. This protects against known vulnerabilities that attackers could exploit.

Purpose: Automatic software updates patch security vulnerabilities as soon as fixes are available, preventing attackers from exploiting known weaknesses.

Why automatic updates are important:

- **Speed:** When a vulnerability is discovered, attackers try to exploit it quickly — automatic updates apply patches before this can happen
- **User behaviour:** Many users ignore or delay manual updates
- **Consistency:** Ensures all systems are protected to the same level
- **Convenience:** No user action required to stay protected

Examples of automatic updates:

- **Operating system updates:** Windows Update, macOS updates
- **Browser updates:** Chrome, Firefox, Edge update automatically
- **App updates:** Mobile apps updating through app stores
- **Antivirus updates:** Virus definitions updating automatically

Advantages of automatic updates

- No user action required
- Patches applied quickly
- Protects against known vulnerabilities
- Ensures consistent protection

Disadvantages of automatic updates

- Updates may cause compatibility issues
- Uses bandwidth and storage
- May restart at inconvenient times
- Users have less control

Worked Example 3 — Biometric measures

Question: Explain how biometric measures can be used to protect a mobile device. [2 marks]

Model answer:
Biometric measures use unique physical characteristics such as fingerprints or facial features to verify identity **(1)**. The device will only unlock if the biometric data matches the registered user, preventing unauthorised access **(1)**.

Exam tip: Explain both WHAT biometrics are and HOW they provide protection. Link to the specific context (mobile devices) if mentioned in the question.

Worked Example 4 — CAPTCHA purpose

Question: Explain what CAPTCHA is and why it is used. [2 marks]

Model answer:
CAPTCHA is a test designed to distinguish between human users and automated bots **(1)**. It is used to prevent automated attacks such as brute force password attempts or spam account creation **(1)**.

Exam tip: Remember CAPTCHA tests if you're human — it doesn't check WHO you are (that's authentication).

Worked Example 5 – Email confirmations

Question: Describe how email confirmations can be used to verify a user's identity. [2 marks]

Model answer:
A link or code is sent to the user's registered email address **(1)**. The user must click the link or enter the code to prove they have access to that email account, confirming their identity **(1)**.

Exam tip: The key point is that it proves the user has access to the email address — if they can receive and respond to the email, they are likely to be the genuine account owner.

Worked Example 6 – Automatic updates

Question: Explain why automatic software updates are important for security. [2 marks]

Model answer:
Automatic updates ensure security patches are installed as soon as they are available **(1)**, preventing attackers from exploiting known vulnerabilities before the user manually updates **(1)**.

Exam tip: Emphasise the speed — automatic updates close vulnerabilities before attackers can exploit them. Many users delay manual updates, leaving systems vulnerable.

3. Protecting against specific threats

Different threats require different prevention methods. This section shows how the five prevention methods can be applied to protect against common cyber security threats.

3.1 Protecting against social engineering

Social engineering attacks manipulate people into giving up confidential information. The best protection combines technical measures with user awareness.

Protection strategies:

- **Email confirmations:** Verify any requests for sensitive information through a separate channel
- **Biometrics:** Protect against shoulder surfing — no password to observe
- **User training:** Teach users to recognise phishing and blagging attempts
- **Verification procedures:** Always verify identity before sharing sensitive information

3.2 Protecting against malware

Malware is malicious software designed to damage systems or steal data. Protection involves both preventing infection and limiting damage.

Protection strategies:

- **Automatic software updates:** Patch vulnerabilities that malware exploits
- **Strong passwords:** Prevent malware from gaining access through weak credentials
- **Anti-malware software:** Detect and remove malicious programs
- **User awareness:** Avoid downloading suspicious files or clicking unknown links

3.3 Protecting against weak passwords

Weak and default passwords are one of the most common security vulnerabilities. Multiple layers of protection can address this threat.

Protection strategies:

- **Password policies:** Enforce minimum length and complexity requirements
- **Biometrics:** Replace or supplement passwords with biometric authentication
- **CAPTCHA:** Prevent brute force attacks on passwords
- **Account lockout:** Lock accounts after several failed login attempts

3.4 Protecting against unpatched software

Unpatched and outdated software contains known vulnerabilities that attackers can exploit.

Protection strategies:

- **Automatic software updates:** Install patches as soon as they are available
- **Update policies:** Ensure all systems are kept up to date
- **Software inventory:** Track what software is installed and its version
- **End-of-life planning:** Replace software that no longer receives updates

3.5 Choosing the right prevention method

Different situations require different security approaches. The table below summarises when each method is most appropriate:

Scenario	Best method(s)	Reason
Unlocking a smartphone	Biometrics	Quick, convenient, secure on small screens

Scenario	Best method(s)	Reason
Creating a new online account	Password + Email confirmation + CAPTCHA	Password for access, email confirms identity, CAPTCHA blocks fake accounts
Online banking login	Password + Biometrics	Two layers of verification for sensitive data
Preventing spam on a contact form	CAPTCHA	Stops automated bots submitting forms
Resetting a forgotten password	Email confirmation	Verifies the person owns the account
Protecting against known vulnerabilities	Automatic updates	Patches security holes quickly
School computer login	Password (with policy)	Works on any device; easy to manage centrally

3.6 Combining prevention methods

The most effective security uses **multiple layers** of protection. If one layer fails, others are still in place.

Example: Protecting an online banking system

- **Password system:** Initial login with strong password
- **CAPTCHA:** Prevents automated login attempts
- **Email confirmation:** Verifies unusual transactions or login locations
- **Biometrics:** Mobile app uses fingerprint for quick, secure access
- **Automatic updates:** Banking app updates automatically with security patches

Together, these five methods provide defence in depth — an attacker would need to bypass all of them to gain access.

Worked Example 7 — Biometrics advantage

Question: State one advantage of using biometric measures instead of passwords. [1 mark]

Model answer:
Biometric data cannot be forgotten, unlike passwords **(1)**.

Exam tip: Other valid answers: cannot be guessed; cannot be observed (shoulder surfing); difficult to steal; quicker to use.

Worked Example 8 — CAPTCHA disadvantage

Question: State one disadvantage of using CAPTCHA. [1 mark]

Model answer:
CAPTCHA can be frustrating or difficult for some users to complete **(1)**.

Exam tip: Other valid answers: accessibility issues for users with visual impairments; takes extra time; advanced bots can sometimes solve them.

Worked Example 9 — Password policies

Question: Describe two features of a strong password policy. [2 marks]

Model answer:
1. Requiring a minimum password length (e.g., at least 8 characters) **(1)**
2. Requiring a mix of character types such as uppercase, lowercase, numbers, and symbols **(1)**

Exam tip: Other valid features: password history (preventing reuse); regular password changes; account lockout after failed attempts.

Worked Example 10 — Combining methods

Question: Explain how biometrics and CAPTCHA can work together to protect a system. [2 marks]

Model answer:
Biometrics verifies the identity of the user by checking their unique physical characteristics **(1)**. CAPTCHA prevents automated bots from attempting to access the system, so both human identity and human presence are verified **(1)**.

Exam tip: Explain what each method does, then explain the combined benefit. They protect against different threats: biometrics stops imposters, CAPTCHA stops bots.

Extended response practice. AQA exams include extended response questions on cyber security. Here is an example with a model banded answer.

Extended Response — Prevention methods in context

Question: A social media company wants to improve security for its users. The company is considering implementing biometric measures, CAPTCHA, and email confirmations.
Discuss how these three methods could be used to protect user accounts. [6 marks]

Model answer (Level 3, 5—6 marks):
The social media company could use all three methods to create multiple layers of protection for user accounts.
Biometric measures could be used in the mobile app to allow users to log in using fingerprint or facial recognition. This is convenient for users as they don't need to type a password on a small screen, and it's more secure because biometric data cannot be guessed, observed, or easily shared with others. If a user's phone is stolen, the thief would not be able to access the account without the correct fingerprint or face.
CAPTCHA could be used during account registration and login to prevent automated attacks. Bots often try to create fake accounts for spam or attempt to break into accounts using brute force attacks. CAPTCHA tests prove that a human is using the system, blocking these automated threats. This protects all users from spam and fake profiles.
Email confirmations could be used to verify new account registrations and to confirm suspicious activities. When a user registers, they would receive an email with a verification link to prove they own that email address. If a login attempt comes from an unusual location or device, the system could send an email asking the user to confirm it was them. This prevents attackers from creating fake accounts or taking over existing accounts.
Together, these three methods address different threats: biometrics verifies the user's identity, CAPTCHA blocks automated attacks, and email confirmations verify ownership and detect suspicious activity. This layered approach means that even if one method is bypassed, the others still provide protection.

Exam tip (6-mark questions): To reach Level 3 (5—6 marks), you must discuss ALL THREE methods mentioned in the question, explain how EACH protects the system, link your points to the specific context (social media), and explain how the methods work TOGETHER. Generic explanations without context will only reach Level 1—2.

Worked Example 11 — Automatic updates context

Question: A company has disabled automatic updates on its computers. Explain one security risk this creates. [2 marks]

Model answer:

Security vulnerabilities will not be patched automatically **(1)**, leaving systems exposed to attacks that exploit known weaknesses until someone manually installs the updates **(1)**.

Exam tip: Emphasise the time gap — between a vulnerability being discovered and a patch being applied, attackers can exploit the weakness.

Worked Example 12 – Biometrics for mobile

Question: Explain why biometric measures are particularly suitable for mobile devices. [2 marks]

Model answer:
Mobile devices have small screens that make typing passwords difficult and slow **(1)**. Biometrics such as fingerprint scanning allow quick, convenient access while maintaining security **(1)**.

Exam tip: The specification specifically mentions biometrics for mobile devices — this is a common exam question. Focus on convenience AND security.

Worked examples summary

This page contains 12 worked examples plus one extended response, covering all the key skills for cyber security prevention. Use this table to find specific types of questions for revision.

Example	Topic	Question type	Marks
1	Defining cyber security	Define	2
2	Purposes of cyber security	State	2
3	Biometric measures for mobile	Explain	2
4	CAPTCHA purpose	Explain	2
5	Email confirmations	Describe	2
6	Automatic updates importance	Explain	2
7	Biometrics advantage	State	1
8	CAPTCHA disadvantage	State	1
9	Password policy features	Describe	2

Example	Topic	Question type	Marks
10	Combining methods	Explain	2
Ext.	Prevention methods in context	Discuss (extended response)	6
11	Automatic updates risk	Explain	2
12	Biometrics for mobile suitability	Explain	2

Quick reference

Key definitions (mark-scheme style)

Term	Definition
Cyber security	The processes, practices and technologies designed to protect networks, computers, programs and data from attack, damage or unauthorised access.
Biometric measures	Using unique physical characteristics (fingerprint, face, iris) to verify a person's identity.
CAPTCHA	A test designed to distinguish between human users and automated bots.
Email confirmation	Verifying identity by sending a link or code to a registered email address.
Automatic software update	Installing security patches automatically without requiring user action.
Strong password	A password that is long, complex, unique, and not based on personal information.

Prevention methods — when to use each

Method	Best used when...
Biometrics	Quick, convenient access is needed (especially mobile devices); password theft is a concern
Passwords	Basic authentication is needed; no special hardware available; users need to share access
CAPTCHA	Automated attacks are a threat; account registration; preventing brute force attacks
Email confirmations	Verifying account ownership; confirming suspicious activity; password resets

Method	Best used when...
Automatic updates	Ensuring consistent protection; users unlikely to update manually; critical security patches

Advantages and disadvantages summary

Quick advantages

- **Biometrics:** Cannot be forgotten or guessed
- **Passwords:** No special hardware needed
- **CAPTCHA:** Blocks automated attacks
- **Email confirm:** Verifies account ownership
- **Auto updates:** No user action required

Quick disadvantages

- **Biometrics:** Expensive; can't be changed if compromised
- **Passwords:** Can be forgotten, guessed, or observed
- **CAPTCHA:** Frustrating; accessibility issues
- **Email confirm:** Relies on email being secure
- **Auto updates:** May cause compatibility issues

Typical pitfalls

Pitfall 1 — Incomplete cyber security definition.
Wrong: "Cyber security is protecting computers."
Right: Processes, practices and technologies designed to protect networks, computers, programs and data from attack, damage or unauthorised access.

Pitfall 2 — Confusing CAPTCHA with authentication.
Wrong: "CAPTCHA checks who the user is."
Right: CAPTCHA checks if the user is HUMAN (not a bot). It doesn't verify identity — that's authentication.

Pitfall 3 — Forgetting biometrics for mobile.
Wrong: Generic biometrics answer.
Right: If asked about mobile devices, mention convenience of fingerprint/face ID compared to typing passwords on small screens.

Pitfall 4 — Vague password advice.
Wrong: "Use a strong password."

Right: Be specific: minimum length, mix of characters, avoid personal information, unique for each account.

Pitfall 5 — Missing the "why" for updates.
Wrong: "Updates add new features."
Right: Security updates patch vulnerabilities that attackers could exploit. Speed is important — automatic updates apply patches before attackers can use them.

Pitfall 6 — One-sided advantages/disadvantages.
Wrong: Only giving advantages OR disadvantages when asked for both.
Right: Always give balanced answers when comparing methods. Every method has pros and cons.

Pitfall 7 — Forgetting context in extended responses.
Wrong: Generic explanation of methods.
Right: Link every point to the specific scenario in the question (e.g., "for the social media company..." or "for the bank's customers...").

Pitfall 8 — Not explaining email confirmations properly.
Wrong: "Email confirmations send an email."
Right: A link/code is sent to the registered email; the user must click/enter it to prove they have access to that email account.

Pitfall 9 — Forgetting biometrics can't be changed.
Wrong: Listing only advantages of biometrics.
Right: A key disadvantage is that if biometric data is compromised, it cannot be changed (unlike a password).

Pitfall 10 — Not combining methods in answers.
Wrong: Treating each method in isolation.
Right: When asked about improving security, discuss how multiple methods work TOGETHER for defence in depth.

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